

Maintenance & Reliability Engineering Report — Q1 2026

1. Maintenance Strategy Overview

SteelForge Manufacturing employs a three-tier maintenance strategy: Preventive Maintenance (PM), Predictive Maintenance (PdM), and Reliability-Centered Maintenance (RCM). The Computerized Maintenance Management System (CMMS) — SAP Plant Maintenance module — tracks all work orders, PM schedules, spare parts inventory, and equipment history. Total maintenance spend for Q1 was \$1.87M against a budget of \$2.1M (89% budget utilization).

The ratio of planned to unplanned maintenance improved from 72:28 in Q4 2025 to 81:19 in Q1 2026, exceeding the world-class benchmark of 80:20. Mean Time Between Failures (MTBF) for critical equipment increased 18% to 1,247 hours. Mean Time To Repair (MTTR) decreased 12% to 2.3 hours, reflecting investments in technician training and spare parts availability improvements.

2. Predictive Maintenance Program

The PdM program covers vibration analysis, infrared thermography, oil analysis, and ultrasonic testing across 156 critical assets. Key findings from Q1: Vibration analysis detected early-stage bearing degradation on CNC Line 4 spindle assembly (Asset #CNC-A-004) in January, but the replacement was deferred to the February maintenance window. The bearing failed catastrophically on Feb 8, causing the 36-hour unplanned shutdown. This event prompted a revision of the PdM decision matrix to lower the deferral threshold for spindle-class bearings from ISO 10816 Zone C to Zone B.

Asset ID	Equipment	PdM Method	Finding	Action Taken
CNC-A-004	Mori Seiki NHX5000	Vibration	Bearing ISO Zone C→D	Replaced (after failure)
STAMP-A-002	Komatsu H2F600	Thermography	Motor hot spot 142°C	Rewound motor — PM
WELD-B-007	Fanuc ArcMate 120	Oil analysis	Fe >150ppm	Gearbox rebuild
HT-B-001	Ipsen TurboTreater	Ultrasonic	Wall thinning 0.8mm	Scheduled Q2 repair
COAT-B-003	Nordson powder booth	Thermography	Oven element failing	Replaced element

3. Spare Parts and Inventory Management

The spare parts storeroom holds \$4.2M in inventory across 8,700 SKUs. Inventory turns improved from 1.8x to 2.1x through the implementation of min/max reorder points based on equipment criticality ranking and historical consumption. Obsolete inventory write-off for Q1 was \$127K, primarily related to legacy Mazak control boards replaced during the CNC retrofit program. A consignment agreement was established with SKF for bearing stock, reducing on-hand inventory by \$310K while maintaining 98% service level.

4. Reliability Metrics Dashboard

Metric	Q4 2025	Q1 2026	Target	Trend
MTBF (hours)	1,056	1,247	1,300	↑ Improving
MTTR (hours)	2.6	2.3	2.0	↑ Improving
PM Compliance	88%	94%	95%	↑ Improving
Planned:Unplanned	72:28	81:19	80:20	✓ Target Met
Spare Parts Service Level	94%	98%	97%	✓ Target Met
Maintenance Cost/Unit	\$1.14	\$1.01	\$1.00	↑ Improving